UNIVERSITY OF CAMBRIDGE

NST1
NATURAL SCIENCES TRIPOS Part IB

Friday 7 June 2024 13.30 to 16.30 (*3 hours plus any pre-agreed individual adjustment*)

Paper PHA/WP

PHARMACOLOGY WRITTEN PRACTICAL PAPER

*Answer **all four** questions. Each **question** is worth **20 marks in total**.*

Extended Answer questions (2) and (4) have no word limit

Diagrams or other handwritten material
Where necessary to perform, show your mathematical workings.
Where appropriate (e.g., for formulae, equations, diagrams or mathematical calculations) you may hand write your answers/workings using the Inspera Scan Paper provided. You should not copy and paste images from any other documents.

Please Note
- *Inspera Scan question codes MUST be noted during the exam.*
- *Please ensure that each Inspera Scan page is labelled appropriately, including your Blind Grade Number.*
- *Inspera Scan pages MUST be placed in the Inspera Scan folder at the completion of the exam and left on your desk.*
- *Inspera scan pages will be scanned by administrators after the completion of the exam.*

Stationery requirements
- *Rough work pad*
- *Inspera Scan pages*
- *Inspera Scan Folder*
- *Equation Sheet*

Calculator – students are permitted to use an approved calculator

An equation sheet is included and may be consulted during the exam

Question 1

The successful chemotherapy of tumours can be severely compromised by acquired resistance to a wide range of chemically and functionally distinct cytotoxic drugs, a phenomenon known as multidrug resistance (MDR). An important factor mediating MDR in cancer cells is the overexpression of the multidrug resistance P-glycoprotein ABCB1 on the surface of resistant cells. ABCB1 acts as an ATP-dependent drug efflux pump that reduces the intracellular concentration of cytotoxic agents, thereby conferring the MDR phenotype. ABCB1 is an ATP-binding cassette transporter that contains cytosolic nucleotide-binding domains, where ATP binding and hydrolysis occur to drive drug efflux, and the plasma membrane-embedded domains that form the pathway for exported drugs to the extracellular environment. In the following experiments, three novel inhibitors of ABCB1 activity (Inhibitor 1, 2, and 3) are characterised. These inhibitors might be further developed into chemosensitizers that can overcome ABCB1-mediated multidrug resistance in clinical applications.

In a first set of experiments, the interaction of ABCB1 with anticancer agent vinblastine (mitotic spindle inhibitor) was examined by equilibrium binding and rapid filtration using [^{3}H]-vinblastine and plasma membrane preparations generated from multidrug resistant Chinese ovary hamster CH^{R}B30 cells. The corresponding binding curve and Scatchard plot are shown in **Figure 1**.

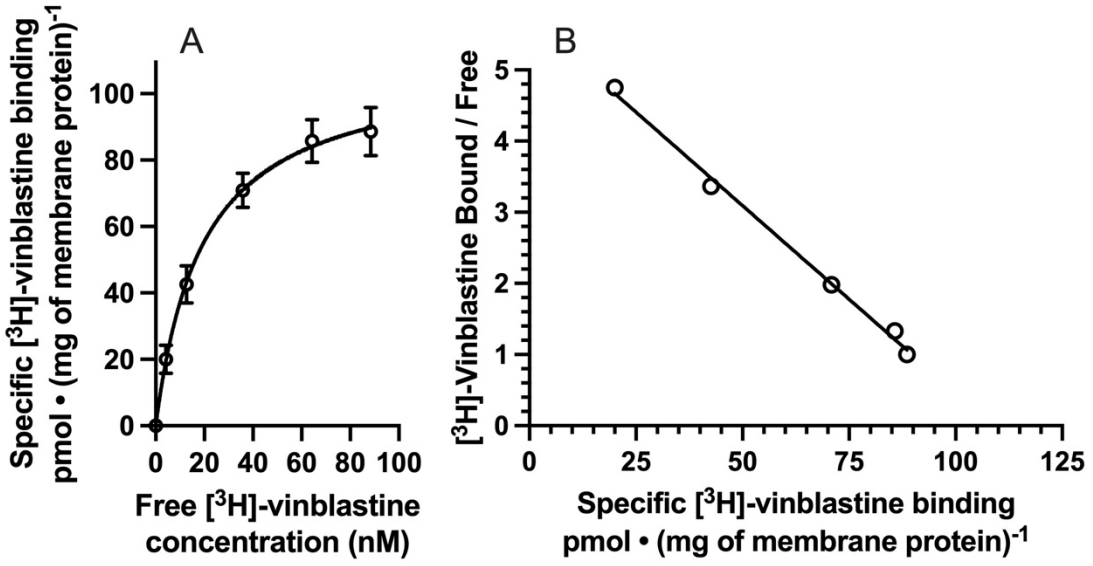

Figure 1. Specific binding of [^{3}H]-vinblastine to ABCB1 in CH^{R}B30 plasma membranes. Membranes (2 μg) were incubated with [^{3}H]-vinblastine (0 – 100 nM) for 2 hours at room temperature after which total binding was measured by rapid filtration. Non-specific binding was determined in the presence of 10 μM unlabelled vinblastine and subtracted from total binding to calculate the specific binding. **Panel A** shows saturable specific binding as a function of the [^{3}H]-vinblastine concentration. **Panel B** show the same data in a Scatchard plot. Data represent observations in four independent experiments (n = 4) with independently prepared batches of plasma membranes of multidrug resistant CH^{R}B30 cells. (pmol = 10^{-12} mol).

[Question 1 continues on the NEXT PAGE]

A. Calculate the binding affinity of ABCB1 for vinblastine. (2 marks)

B. If one vinblastine molecule binds per ABCB1 transporter, 10^6 cells contain 250 μg total protein, and the ratio (w/w) of membrane protein over total protein equals 0.228, calculate the number of ABCB1 transporters per CH^RB30 cell (Avogadro's number equals 6.023 X 10^{23}). Compare your calculations for ABCB1 with the glucose transporter, which is one of the most abundant transporters with roughly 2 X 10^5 transport proteins per cell. Note that the CH^RB30 cell line was generated from a parental drug-sensitive AuxB1 cell line by stepwise selection in cytotoxic colchicine-containing growth medium. (4 marks)

In a second set of experiments, the interaction of the inhibitors with ABCB1 was studied in a [^{3}H]-vinblastine displacement experiment. CH^RB30 plasma membranes were incubated with 12 nM [^{3}H]-vinblastine in the presence of increasing concentrations of unlabelled Inhibitor 1, 2, or 3, which all interact with ABCB1 in a 1:1 ratio. The results are shown in **Figure 2**.

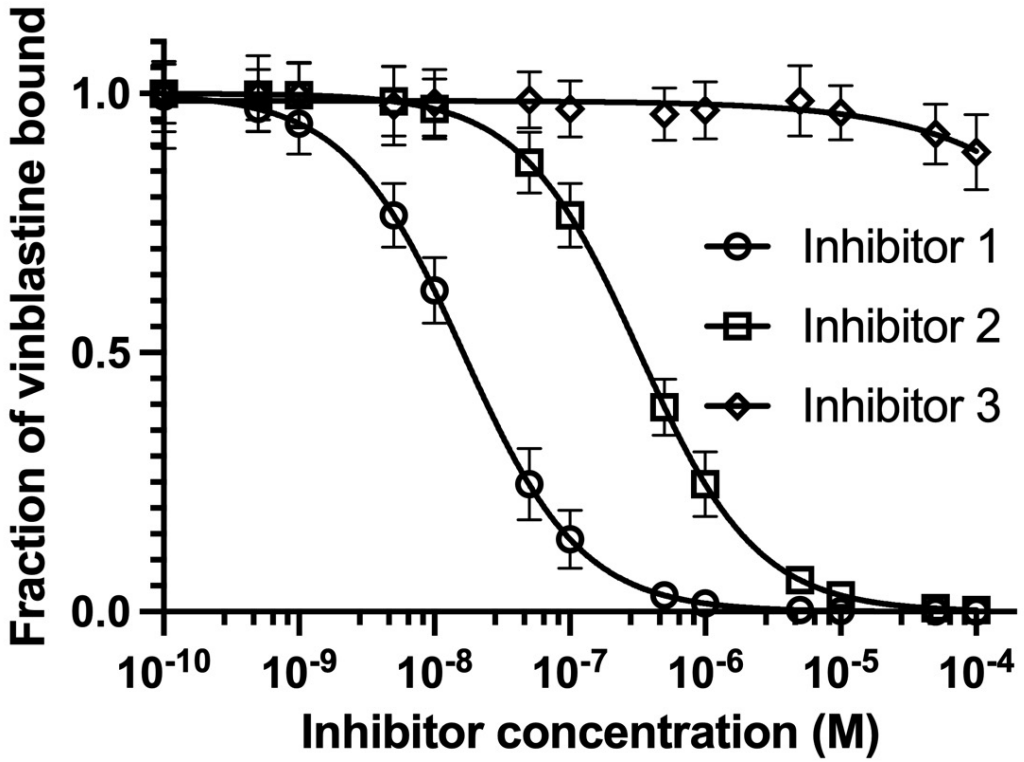

Figure 2. Displacement of [^{3}H]-vinblastine binding to ABCB1 by unlabelled Inhibitor 1, 2, and 3. The specific binding of [^{3}H]-vinblastine to ABCB1 in CH^RB30 plasma membranes was measured as described in Figure 1 at a fixed concentration of 12 nM [^{3}H]-vinblastine and inhibitor concentrations in a range of 10^{-10} to 10^{-4} M. Data represent observations in four independent experiments (n = 4) with independently prepared batches of plasma membranes of multidrug resistant CH^RB30 cells.

[Question 1 continues on the NEXT PAGE]

C. Calculate the binding affinity of ABCB1 for Inhibitor 1 and 2. (2 marks)

The effect of Inhibitor 3 on the ATPase activity of purified ABCB1 was measured at a range of ATP concentrations (10 μM to 20 mM). The results of this third set of experiments are shown in **Figure 3**.

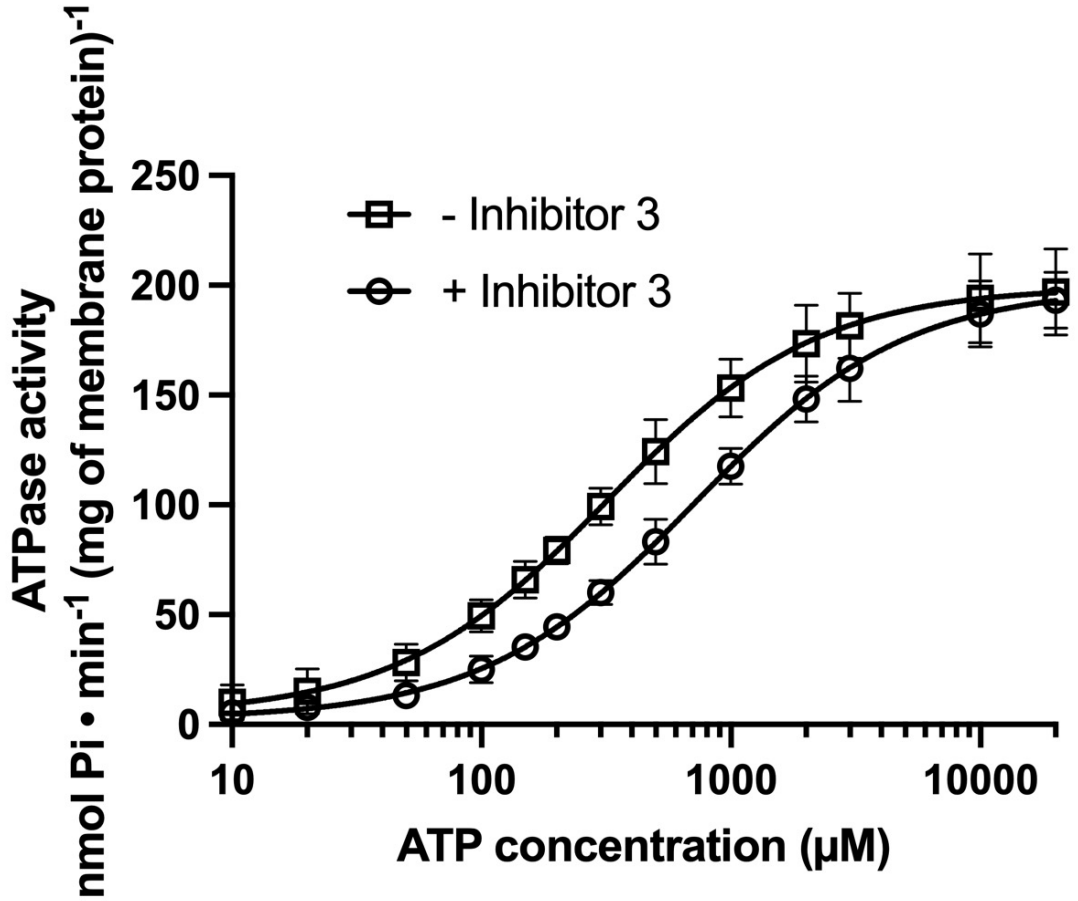

Figure 3. ATPase activity of purified ABCB1 in the absence and presence of 100 nM of Inhibitor 3. Data represent observations in four independent experiments (n = 4) with independently prepared batches of purified protein.

D. Calculate the binding affinity of ABCB1 for Inhibitor 3. (2 marks)

E. Given the key assumptions pertinent to the data in Figure 2 and 3, what can you conclude about the mechanism of Inhibitor 3? (2 marks)

[Question 1 continues on the NEXT PAGE]

In a final set of experiments, the accumulation of vinblastine and inhibitors was measured in CH^{R}B30 cells and the drug-sensitive AuxB1 parental control cells. The results are shown in **Figure 4**.

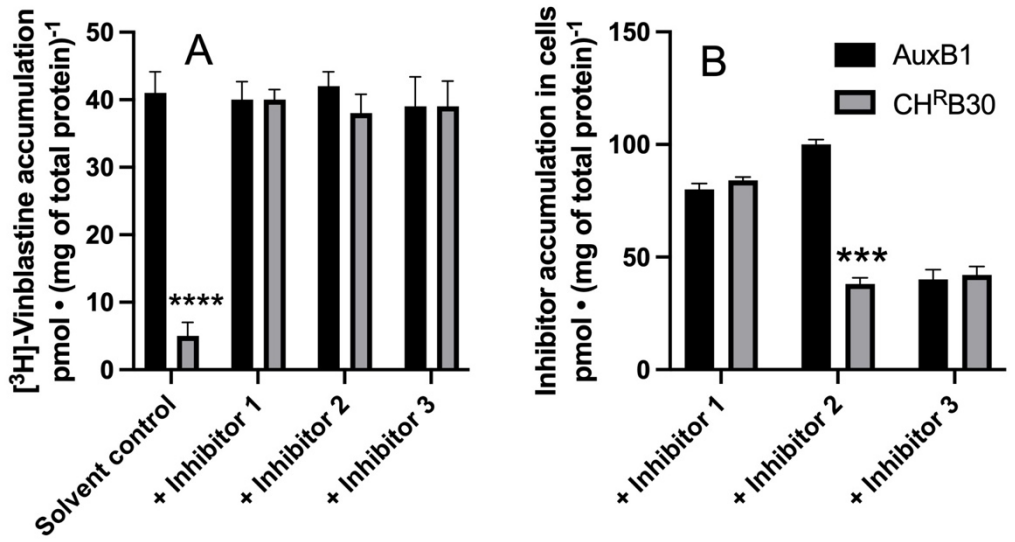

Figure 4. Accumulation of vinblastine and inhibitors in AuxB1 control cells (black bars) and ABCB1-expressing CH^{R}B30 cells (grey bars). In **Panel A**, the accumulation of [^{3}H]-vinblastine was measured in cells over a 60 min period by rapid filtration at 37 °C in α-minimum essential medium (α-MEM) containing 10% foetal calf serum and 100 nM [^{3}H]-vinblastine under a 5% CO_2 atmosphere. These experiments were performed in the absence of inhibitors (solvent control) or in the presence of 0.2 μM of Inhibitor 1, 2, or 3. In **Panel B**, the accumulation of 0.2 μM of Inhibitor 1, 2, or 3 in AuxB1 and CH^{R}B30 cells was measured by mass spectrometry under similar conditions as in Panel A, but in the absence of vinblastine. Data represent observations in four independent experiments (n = 4) with independently prepared batches of cells. Histogram displays the mean ± s.e.m. (one-way analysis of variance; ***P < 0.001, ****P < 0.0001 relative to the AuxB1 control).

F. Based on the total body of data in Figure 1 to 4, compare and contrast the mechanisms of Inhibitor 1, 2, and 3. (4 marks)

G. Which inhibitor has potential in further drug development? What further information and experiments would support your choice? (4 marks)

End of Question 1

2. Outline the principal intracellular signalling mechanisms used by membrane receptors. (20 marks)

[Question 3 starts on the NEXT PAGE]

Question 3

An investigational antagonist, named CA-24™, of the chemokine CXCR4 receptor was evaluated in a Phase I clinical trial for its safety and basic pharmacokinetic profiling in healthy adult volunteers. **Figure 1** shows exemplar plasma concentration versus time data obtained from a healthy volunteer weighing 80 kg who received a single dose of CA-24™ (25 $\mu g.kg^{-1}$) either intravenously or orally (with no prior meal). The plasma concentrations are also listed in **Table 1**.

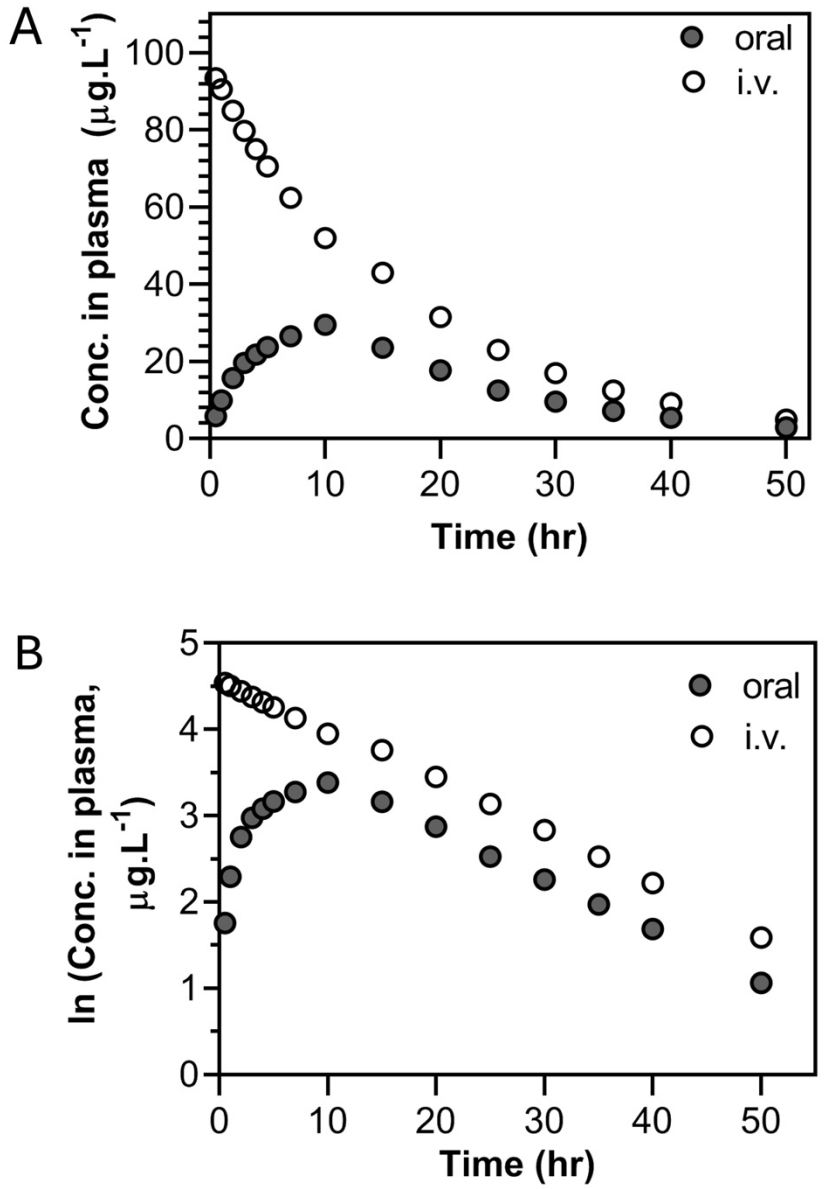

Figure 1. Plasma concentration versus time of CA-24™ in a healthy adult volunteer. **Panel A** shows the linear plot and **Panel B** shows the natural log of plasma concentrations plotted against time. Data points collected from oral and intravenous (i.v.) administration of the drug are shown in grey and open circles, respectively.

[Question 3 continues on the NEXT PAGE]

Table 1. Plasma concentrations of CA-24™ measured at various times after oral or i.v. administration of a single dose (25 μg.kg^{-1}).

Time (hr)	Concentration in plasma after oral administration (μg.L^{-1})	Concentration in plasma after i.v. administration (μg.L^{-1})
0.5	5.8	93.4
1	9.9	90.5
2	15.7	85
3	19.6	79.8
4	21.8	75
5	23.7	70.5
7	26.5	62.4
10	29.5	52
15	23.6	43
20	17.7	31.5
25	12.5	23
30	9.6	17
35	7.2	12.5
40	5.4	9.2
50	2.9	4.9

A. Comment on the plots shown in Figure 1. (2 marks)

B. From these data, estimate the half-life for elimination, clearance, bioavailability, and volume of distribution of CA-24. (8 marks)

After this initial assessment, the drug was administered intravenously to the same patient at a dose of 12.5 μg.kg^{-1} every 9 hours.

C. (i) Calculate the maximum and minimum plasma concentrations. (3 marks)

(ii) The minimum effective concentration of CA-24 is 70 μg.L^{-1}. If it is administered as a constant infusion, what dose rate would be required to achieve therapeutic benefit within 5 hours. (3 marks)

[Question 3 continues on the NEXT PAGE]

In a follow-up study to assess drug administration and patient compliance, the same volunteer took 2 mg of CA-24™ orally on three separate days with an interval of one week in between each. In Week 1, the volunteer had not eaten for at least 2 hours prior to administration of the drug and continued to fast for at least 6 hours after. In Week 2, the drug was taken together with a main meal containing protein and carbohydrates. In Week 3, the volunteer had a glass of milk, 5 min prior to taking the drug.

In each case, the plasma concentrations of CA-24™ were determined and plotted against time (**Figure 2**).

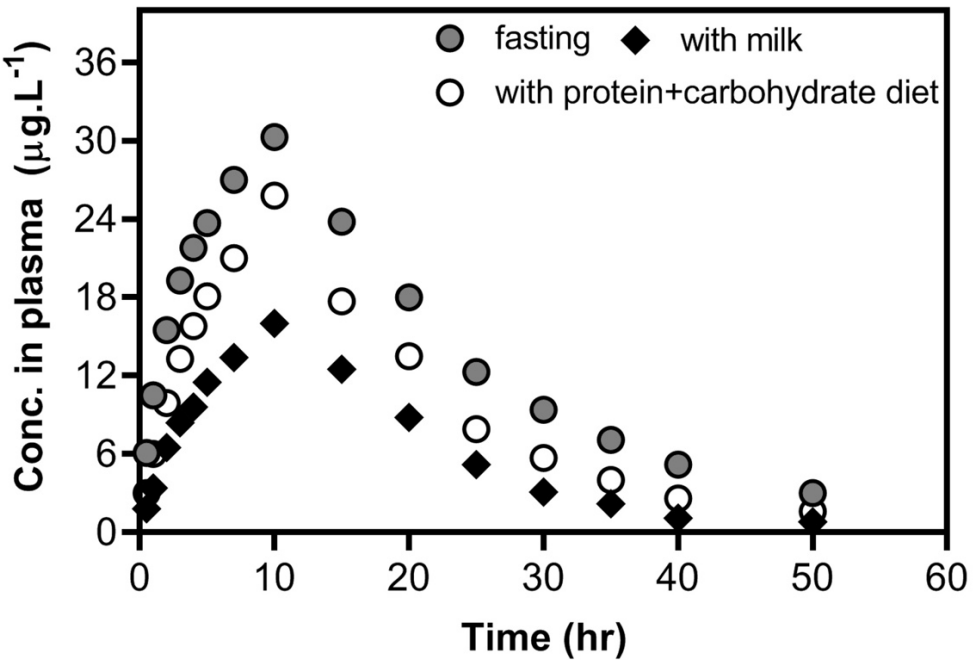

Figure 2. Plasma concentration versus time profiles of CA-24™ in a healthy adult volunteer under different conditions studied in three different weeks.

D. Comment on the data shown in Figure 2. (4 marks)

End of Question 3

4. Describe the factors that influence the oral bioavailability of drugs. (20 marks)

END OF PAPER

PHARMACOLOGY EQUATION SHEET

The following represent a core set of equations from NST 1B PHARMACOLOGY course that may be used in conjunction with your understanding of the concepts described in the course.

LIGAND BINDING EQUATIONS

One ligand

$$\alpha = \frac{K_A[D]}{1 + K_A[D]}$$

Two ligands

$$\alpha = \frac{K_1[D_1]}{1 + K_1[D_1] + K_2[D_2]} \; (for\ ligand\ 1)$$

Dose Ratio

$$\frac{[D']}{[D]} = \text{dose ratio} = 1 + K_{Antagonist}[A]$$

Inhibition of radioligand binding

$$K_U = \frac{1 + [L]K_L}{IC_{50}}$$

PHARMACOKINETICS EQUATIONS

Single intravenous dose, one compartment model

$$C = C_0 . e^{-k_e . t} \qquad t_{0.5} = \frac{0.693}{k_e} \qquad AUC = \frac{C_0}{k_e} \qquad CL = \frac{Dose}{AUC} \qquad V_d = \frac{CL}{k_e}$$

Single oral dose

$$F = \frac{AUC_{oral}}{AUC_{i.v.}} \times \frac{Dose_{i.v.}}{Dose_{oral}} \qquad CL = \frac{Dose . F}{AUC_{total}}$$

Constant infusion

$$C_{ss} = \frac{R_{in}}{CL} \qquad t = -\frac{1}{k_e} \ln\left[1 - \frac{C_t}{C_{ss}}\right]$$

Multiple intravenous doses

$$C_{av,\ ss} = \frac{AUC_{ss}}{\tau} = \frac{dose}{CL . \tau} \qquad C_{max,\ ss} = \frac{dose}{V_d}\left(\frac{1}{1 - e^{-k_e . \tau}}\right) \qquad C_{min,\ ss} = C_{max,\ ss} . e^{-k_e . \tau}$$